The Confluent Hypergeometric Functions

Theory, Computation and Applications

Polynomial Physicists

Aman Kumar Md Farhan Arani Ghosh
Abhilasha Shashank

PH-511: Mathematical Physics
IIT Mandi

November 2026

Presentation Roadmap

1. Introduction and Kummer's differential equation
2. Series solution and integral representations
3. Identities, asymptotics and special cases
4. Worked examples
5. Hydrogen atom application
6. Numerical implementation and verification
7. Current research applications and conclusion

Introduction: Kummer's Equation

Many problems in mathematical physics lead to

$$\boxed{zy''(z) + (b - z)y'(z) - ay(z) = 0.}$$

This is Kummer's equation, also called the confluent hypergeometric equation.

- The solution regular at $z = 0$ is $M(a, b, z) = {}_1F_1(a; b; z)$.
- A second independent solution is Tricomi's function $U(a, b, z)$.
- It is obtained from the Gauss hypergeometric equation when two singular points merge (confluence).

Series Solution: Frobenius Method

Assume $y(z) = \sum_{k=0}^{\infty} c_k z^{k+s}$. The indicial equation is

$$s(s+b-1) = 0, \qquad s = 0 \text{ or } s = 1-b.$$

For the regular solution $s = 0$,

$$c_{k+1} = \frac{a+k}{(b+k)(k+1)} c_k.$$

Taking $c_0 = 1$ gives

$$\boxed{M(a,b,z) = \sum_{k=0}^{\infty} \frac{(a)_k}{(b)_k} \frac{z^k}{k!}.}$$

Here $(a)_k = a(a+1)\cdots(a+k-1)$ is the Pochhammer symbol.

Convergence and Polynomial Termination

The series converges for every finite z, provided b is not $0, -1, -2, \ldots$.

Polynomial condition

If $a = -m$, where $m = 0, 1, 2, \ldots$, then $(-m)_k = 0$ for $k > m$. The series terminates and becomes a polynomial.

For example,

$$\boxed{M(-1, 2, z) = 1 - \frac{z}{2}.}$$

This termination property is essential in the hydrogen-atom bound-state problem.

Integral Representations

For $\mathrm{Re}(b) > \mathrm{Re}(a) > 0$, Euler's integral is

$$M(a, b, z) = \frac{\Gamma(b)}{\Gamma(a)\Gamma(b-a)} \int_0^1 e^{zt} t^{a-1} (1-t)^{b-a-1} \, dt.$$

For $\mathrm{Re}(a) > 0$ and $\mathrm{Re}(z) > 0$,

$$U(a, b, z) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-zt} t^{a-1} (1+t)^{b-a-1} \, dt.$$

These representations connect the functions with Gamma and Beta integrals and offer alternative numerical evaluation methods.

Important Identities

Kummer's transformation

$$\boxed{M(a,b,z) = e^z M(b-a,b,-z).}$$

Derivative identity

$$\boxed{\frac{d}{dz}M(a,b,z) = \frac{a}{b}M(a+1,b+1,z).}$$

Contiguous relation

$$bM(a,b,z) - bM(a-1,b,z) - zM(a,b+1,z) = 0.$$

These identities help derive related formulas and can improve numerical evaluation, especially for negative arguments.

Asymptotic Behaviour

For large positive real z, the dominant behaviour is

$$M(a,b,z) \sim \frac{\Gamma(b)}{\Gamma(a)} e^z z^{a-b}, \qquad z \to +\infty.$$

For large negative real z, the leading algebraic behaviour is

$$M(a,b,z) \sim \frac{\Gamma(b)}{\Gamma(b-a)} (-z)^{-a}, \qquad z \to -\infty.$$

- On the positive real axis, the function generally grows exponentially.
- On the negative real axis, the leading behaviour is a power law.
- If a is a non-positive integer, the series terminates and gives a polynomial.

Special Functions as Special Cases

Function	Representation
Exponential	$M(a, a, z) = e^z$
Elementary case	$M(1, 2, z) = \dfrac{e^z - 1}{z}$
Error function	$\operatorname{erf}(z) = \dfrac{2z}{\sqrt{\pi}} M\left(\frac{1}{2}, \frac{3}{2}, -z^2\right)$
Laguerre polynomial	$L_n^{(\alpha)}(z) = \dfrac{(\alpha + 1)_n}{n!} M(-n, \alpha + 1, z)$

The confluent hypergeometric function is a building block for many familiar special functions.

Worked Examples

1. **Polynomial solution:** For $a = -1$, $b = 2$,

$$M(-1, 2, z) = 1 - \frac{z}{2}.$$

2. **Exponential special case:** If $a = b$, then $(a)_k/(b)_k = 1$, so

$$M(a, a, z) = \sum_{k=0}^{\infty} \frac{z^k}{k!} = e^z.$$

3. **Elementary closed form:**

$$M(1, 2, z) = 1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots = \frac{e^z - 1}{z},$$

with continuous value 1 at $z = 0$.

Application to the Hydrogen Atom

In atomic units, the radial Schrödinger equation is

$$u'' + \left[\frac{2}{r} - \frac{l(l+1)}{r^2} + 2E\right] u = 0.$$

For bound states, set

$$E = -\frac{1}{2\nu^2}, \quad \rho = \frac{2r}{\nu}, \quad u = \rho^{l+1} e^{-\rho/2} v(\rho).$$

The equation for v becomes Kummer's equation with

$$a = l + 1 - \nu, \qquad b = 2l + 2.$$

The regular solution is $v(\rho) = M(l + 1 - \nu, 2l + 2, \rho)$.

Hydrogen Atom: Normalisability and Quantisation

A bound-state wave function must be normalisable. The exponentially growing contribution must be excluded, requiring the Kummer series to terminate:

$$a = l + 1 - \nu = -n_r, \qquad n_r = 0, 1, 2, \ldots$$

Therefore,

$$\nu = n = n_r + l + 1.$$

Allowed hydrogen energy levels

$$\boxed{E_n = -\frac{1}{2n^2}, \qquad n = 1, 2, 3, \ldots}$$

Normalisability $\Rightarrow$ Series termination $\Rightarrow$ Discrete energy levels

Numerical Implementation and Verification

The project used Python tools:

- **SymPy:** symbolic series and differential-equation checks.
- **mpmath:** arbitrary-precision numerical evaluation.
- **SciPy:** reference evaluation using `hyp1f1`.
- **Matplotlib:** numerical plots and visual comparisons.

Checks included:

- Substitution of a truncated series into Kummer's equation.
- Numerical verification of Kummer's transformation.
- Comparison of Euler integral, manual series and SciPy values.
- Comparison of numerical ODE solutions with the analytical function.

Numerical Results and Computation

The report presents comparisons of:

1. Analytical Kummer function and numerical RK45 solution.
2. Absolute error for different solver tolerances.
3. Euler integral, series recurrence and SciPy evaluation.
4. Series convergence as more terms are retained.
5. Hydrogen radial wave functions, radial probability distributions and energy levels.
6. Positive- and negative-argument asymptotic behaviour.

Numerical caution

For large negative z, alternating large terms can cause cancellation and loss of significant digits. Kummer's transformation and arbitrary-precision arithmetic help cross-check results.

Applications in Current Research

Quantum mechanics and atomic physics

- Coulomb bound states and scattering; hydrogen radial functions.
- Harmonic oscillator, Morse potential and exactly solvable models.

Wireless communications and statistics

- Appears in expressions related to Nakagami-m, non-central chi-square and Rician models.

Gravitational and field-theoretic problems

- Perturbation equations on curved backgrounds and strong-field particle-production problems.

Mathematical methods

- Related to generalised hypergeometric functions, Feynman integrals and orthogonal-polynomial theory.

Conclusion

- Kummer's function is the regular solution of the confluent hypergeometric differential equation.
- Its series, integral representations, identities and asymptotics provide different ways to study and compute it.
- Many familiar special functions occur as special cases.
- In hydrogen, series termination enforces normalisability and gives the discrete energy spectrum.
- Symbolic and numerical tools provide independent verification.

Thank You!

Questions?

Team Roles and References

Team responsibilities

- Aman Kumar: Literature review and theory.
- Md Farhan: Introduction, series and integral representation.
- Abhilasha: Software implementation, verification and plots.
- Arani Ghosh: Current research applications, bibliography and typesetting.
- Shashank: Reading, discussion and proofreading.

Selected references


M. Abramowitz and I. A. Stegun (eds.), *Handbook of Mathematical Functions*, Chapter 13, Dover, 1972.


NIST Digital Library of Mathematical Functions, Chapter 13, `https://dlmf.nist.gov/13`.


G. B. Arfken, H. J. Weber and F. E. Harris, *Mathematical Methods for Physicists*, 7th ed., 2012.